Cross-Architecture KV Cache Quantization: Why No Single Method Wins

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Abstract

KV cache quantization is critical for deploying large language models on resource-constrained devices. We present the first systematic cross-architecture perplexity evaluation of KV cache quantization methods across three fundamentally different attention designs: MQA (Gemma 4 E2B, 1 KV head), wide-GQA (Grouped Query Attention; Mistral 7B, 8 KV heads), and narrow-GQA (Qwen2-7B, 4 KV heads).

Our key finding is that **no single method dominates across architectures**. KIVI wins 6 of 9 cells in our v2 evaluation matrix, but catastrophically fails on Gemma MQA at 2-bit (PPL 470 vs FP16 28, i.e., >16× degradation—effectively unusable). Architecture dependence is the dominant signal: the "right" method depends on the KV head count, not the bit width.

Building on this finding, we introduce **FajarQuant v3.1**, an adaptive per-head method selection system that profiles each KV head's statistical properties and routes to the optimal quantization strategy. V3.1 matches or beats the best fixed method in **7 of 9** evaluation cells, achieves zero catastrophic failures (vs 2 in the non-adaptive v3.0), and discovers architecture-specific optima: **−24%** PPL vs KIVI on Gemma MQA at 3-bit via PCA rotation, and **−35%** vs KIVI on Qwen2 GQA at 2-bit via PPL-guided strategy selection.

We additionally contribute: (1) a canonical R-α.1 model-surgery evaluation protocol enabling fair cross-method comparison; (2) **FajarQuant v2**, an outlier-aware calibrated PCA method that improves 63–81% over v1 at 2-bit; and (3) a native Fajar Lang implementation with compile-time safety guarantees (`@device` context, SE023 type rule) running 5× faster than the Python reference. V3.1 additionally introduces per-head PCA calibration for GQA models and PPL-guided strategy selection at 2-bit.

1 Introduction

1.1 The KV Cache Memory Problem

Transformer-based large language models require storing key-value (KV) caches for autoregressive generation. For a model with L layers, H heads, dimension d per head, and sequence length N:

$$\text{Memory} = 2 \times L \times H \times N \times d \times \text{sizeof(dtype)} \quad (1)$$

For a 7B model (L=32, H=32, d=128) at N=4096 in FP16: **4 GB**. Embedded targets typically have 256 KB–8 MB SRAM. Aggressive quantization is essential.

1.2 Prior Work

TurboQuant [Zandieh et al., 2025] applies random orthogonal rotation followed by coordinate-wise Lloyd-Max quantization, achieving MSE within 2.7× of the information-theoretic optimum. **KIVI** [Liu et al., 2024] observes that keys have large magnitude variation across channels and values across tokens, and applies per-channel key and per-token value quantization accordingly.

1.3 Contributions

We identify three opportunities to improve upon prior work:

1. **The rotation is data-agnostic.** KV cache vectors exhibit moderate low-rank structure (SV1/SV2 ratio 2.73 on Gemma 4). PCA rotation can exploit this.
2. **Dequantization wastes memory.** Standard attention requires fully dequantizing each key, allocating $O(N \cdot d)$ extra memory.
3. **Flat bit allocation wastes budget.** Recent tokens receive far more attention weight than old tokens.

All three are combined in FajarQuant v2 (§8), and the adaptive per-head selection system (v3.1, §16) builds on the cross-architecture findings to route each KV head to its optimal quantization strategy.

2 Background: TurboQuant

Lemma 1 (Beta Distribution after Rotation)**.** *For x on the unit sphere $\mathbb{S}^{d-1}$, after orthogonal rotation $y = \Pi x$, each coordinate y_i follows Beta$(\frac{d-1}{2}, \frac{d-1}{2})$ on $[-1, 1]$.*

Theorem 2 (TurboQuant Distortion Bound)**.**

$$D_{mse}(b) \leq \frac{\sqrt{3}\pi}{2} \cdot \frac{1}{4^b} \leq 2.72 \cdot D^*(b) \tag{2}$$

3 Evaluation Methodology

3.1 Canonical Perplexity Protocol (R-α.1)

Prior KV cache quantization papers use varying evaluation protocols, making cross-method comparison unreliable. We adopt the **R-α.1 model surgery** protocol, following KVQuant [**?**] and SKVQ [**?**]:

1. Load the full-precision model.
2. Patch the attention forward pass to apply quantization *after* key/value projection but *before* the attention score computation. This is "model surgery"—the KV cache is quantized in-flight, not post-hoc.
3. Evaluate perplexity on WikiText-2 test using non-overlapping chunks of length L=2048, computing cross-entropy on positions $[1, L)$ per chunk.
4. Report perplexity as exp(mean NLL) over n=30 chunks (61,410 tokens).

The FP16 baseline is produced by the same forward pass with quantization disabled (identity patch). This ensures the baseline reflects the model's actual WikiText-2 perplexity, not an artifact of the evaluation harness.

3.2 The v1 → v2 Transition

FajarQuant v1 used PCA rotation calibrated on key data, with symmetric per-tensor quantization. Under the original non-canonical evaluation protocol (prefix+target post-hoc cache mutation), v1 appeared to outperform all baselines at 2–3 bit on Gemma 4 E2B. When we switched to the canonical R-α.1 protocol, v1's advantage disappeared: **v1 lost to TurboQuant by 5.6**× on Gemma 2-bit (PPL 125 vs PPL 40).

Diagnosis revealed two root causes: (1) v1's per-tensor symmetric quantization destroyed outlier channels that carry disproportionate information in real KV cache; (2) the PCA rotation without outlier handling amplified rather than reduced quantization error on high-kurtosis distributions.

FajarQuant v2 addresses both: it adds **15% outlier extraction** (top channels preserved in higher precision) and uses a **fused Hadamard+quantize kernel** that spreads outliers before quantization. V2 improves over v1 by 63–81% at 2-bit across all three models (Table 5).

4 Innovation 1: Adaptive PCA Rotation

4.1 Motivation

Random rotation uniformizes coordinate distributions—optimal for worst-case data but suboptimal for structured KV cache vectors. We analyze real KV cache from Google Gemma 4 E2B and find moderate low-rank structure: the mean singular value ratio $\sigma_1/\sigma_2 = 2.73 \pm 0.56$ across 150 layer-head samples, with Layer 1 showing the strongest structure ($\sigma_1/\sigma_2 = 4.04$). The top-3 singular values capture 66.7% of total energy in keys—significantly more than the 50% expected from random data.

4.2 Method

Given calibration data $\{x_1, \ldots, x_N\}$, we compute the covariance $C = \frac{1}{N}\sum_i x_i x_i^T$ and extract eigenvectors via eigendecomposition to form R_{pca}. After PCA rotation, each coordinate corresponds to a principal component with distinct variance. We then apply per-coordinate uniform quantization with independent scale and zero-point per dimension—allocating quantization resolution proportional to each component's dynamic range.

Theorem 3 (Adaptive Rotation Bound)**.** *Let C have eigenvalues $\lambda_1 \geq \cdots \geq \lambda_d$. With per-coordinate quantization after PCA rotation:*

$$MSE(R_{pca}) \leq MSE(R_{rand}) \tag{3}$$

4.3 Results on Real Gemma 4 Data

We evaluate on KV cache extracted from Google Gemma 4 E2B (35 layers, 1 KV head, d=256) using 50 diverse prompts from WikiText-2. All methods use per-coordinate quantization for fair comparison.

FajarQuant consistently achieves the lowest key MSE at all bit widths, with 4–6% improvement over both baselines. The improvement is modest but *consistent*—PCA rotation provides a reliable edge on data with moderate low-rank structure.

Table 1: Key quantization MSE on real Gemma 4 E2B KV cache (d=256, 150 samples)

Bits	FajarQuant	KIVI	TurboQuant
b=2	$\mathbf{8.72 \times 10^{-4}}$	9.31×10^{-4}	9.18×10^{-4}
b=3	$\mathbf{1.60 \times 10^{-4}}$	1.66×10^{-4}	1.67×10^{-4}
b=4	$\mathbf{3.46 \times 10^{-5}}$	3.60×10^{-5}	3.63×10^{-5}
Improvement over baselines:			
b=2	—	+6.3%	+4.9%
b=3	—	+3.8%	+4.3%
b=4	—	+4.0%	+4.8%

Table 2: Memory savings with fused attention (d=128, b=3)

Seq. Length	FP64 KV	Quantized	Fused Extra
1K tokens	2,048 KB	256 KB	64 B
4K tokens	8,192 KB	1,024 KB	64 B
16K tokens	32,768 KB	4,096 KB	64 B

5 Innovation 2: Fused Quantized Attention

Standard attention dequantizes all N key vectors before computing dot products, requiring $O(N \cdot d)$ extra memory. We eliminate this by computing directly on quantized indices via codebook lookup:

$$\text{score}_i = q^T \cdot c[k_{\text{idx}}[i]], \quad \text{output} = \sum_i w_i \cdot c[v_{\text{idx}}[i]] \tag{4}$$

Theorem 4 (Fused Attention Equivalence)**.** *The fused computation produces results identical to standard dequantize-then-attend, with extra memory $O(2^b)$ instead of $O(N \cdot d)$.*

6 Innovation 3: Hierarchical Multi-Resolution

We define a bit schedule assigning more bits to recent tokens:

7 Embedded Deployment via Fajar Lang

FajarQuant v2 is implemented natively in Fajar Lang, a statically-typed systems language with dual-context safety annotations. This is the first KV cache quantization method with compile-time safety guarantees enforced by the language itself.

Table 3: Hierarchical bit budget (N=10,000, d=128)

Tier	Token Age	Bits	Count	Subtotal
1	0–256	4	256	1,024
2	257–1024	3	768	2,304
3	1025–4096	2	3,072	6,144
4	4097+	1	5,904	5,904
Total	—	**1.54 avg**	10,000	**15,376**
Flat 3-bit	—	3.0	10,000	30,000
Savings				**48.7%**

Implementation note. This paper evaluates two distinct implementations: (1) a **Python/PyTorch reference** (`fajarkraton/fajarquant` on GitHub) used for all perplexity and MSE benchmarks in §9–§16, running on an RTX 4090 GPU; and (2) a **native Fajar Lang implementation** (`fajarkraton/fajar-lang`, `src/runtime/ml/`) used for the embedded deployment story in this section and for the performance comparison in §15. A separate kernel-mode port exists in FajarOS (`fajarkraton/fajaros-x86`, `kernel/compute/fajarquant.fj`) demonstrating Ring 0 KV cache quantization for SmolLM-135M inference without userspace—that port is a deployment demonstration, not a benchmark target.

7.1 Language-Level Quantization Types

Fajar Lang provides a first-class `Quantized<T, BITS>` type (SE023 compiler rule). Using a `Quantized` value where a `Tensor` is expected is a compile-time error—preventing accidental use of packed integer data as float values:

Listing 1: SE023: Quantized type safety

```
let q = quantize(kv_row, 4)
// matmul(q, q)  // ERROR SE023: call
    dequantize()
let t = dequantize(q)  // OK
matmul(t, t)            // OK
```

7.2 @device Context Enforcement

The `@device` annotation restricts functions to ML-safe operations: tensor math, quantization, and Hadamard transforms are allowed; raw pointers, IRQ, and heap allocation are compiler errors. The entire FajarQuant v2 pipeline is implementable in `@device`:

Listing 2: Full v2 pipeline in @device context

```
@device
fn fq_v2_encode(kv: Tensor, bits: i64)
```

```
    -> Quantized {
  hadamard_quantize(kv, bits) // fused SIMD
}
@device
fn fq_v2_decode(q: Quantized) -> Tensor {
  hadamard_inverse(dequantize(q))
}
```

7.3 Performance: Native vs Python

The Fajar Lang implementation uses AVX2 SIMD intrinsics for the Hadamard butterfly (1.9× speedup over scalar) and a fused Hadamard+quantize kernel (1.6× over separate passes). End-to-end, the native implementation runs **5.0× faster** than the Python+NumPy reference on an 8-token × 64-dim workload (28 ms vs 142 ms, release build, 10 iterations averaged).

7.4 Zero-Heap KV Cache

The `QuantizedKVCache` struct provides a stack-allocated KV cache with overflow detection. For a 2-bit cache with 2048 max tokens, 12 layers, 9 heads, and 64-dim heads, the effective size is ∼7 MB — fitting in the L3 cache of modern processors (36 MB on i9-14900HX).

8 FajarQuant v2: Outlier-Aware Calibrated PCA

FajarQuant v1's naive per-tensor symmetric quantization after PCA rotation worked well on MSE benchmarks but failed under canonical perplexity evaluation (§3.2). V2 addresses the two root causes with a redesigned pipeline.

8.1 Algorithm

Given input KV cache tensor $X \in \mathbb{R}^{N \times d}$:

1. **Hadamard pre-rotation.** Apply the Walsh-Hadamard Transform H_d to spread outlier magnitudes across channels: $X' = X \cdot H_d/\sqrt{d}$. This is $O(Nd \log d)$ via the fast butterfly algorithm and reduces the dynamic range of each channel.
2. **Outlier extraction.** Identify the top-k channels by absolute magnitude (default $k = \lfloor 0.15 \cdot d \rfloor$, i.e., 15%). These channels are preserved in higher precision (FP16) rather than quantized, because they carry disproportionate information. The remaining $(1-0.15) \cdot d$ channels are quantized.
3. **Per-coordinate symmetric quantization.** Each non-outlier channel i is quantized independently with scale $s_i = \max |X'_{:,i}|/(2^{b-1} - 1)$.
4. **Reconstruction.** Dequantize non-outlier channels, merge with preserved outliers, apply inverse Hadamard: $\hat{X} = \hat{X}' \cdot H_d/\sqrt{d}$.

Table 4: v1 → v2 ablation on Gemma 2-bit (R-α.1, n=30)

Configuration	PPL (2-bit)	Δ vs v1
FQ v1 (PCA only)	125.19	—
+ Hadamard pre-rotation	161.31*	−28.8%
+ 15% outlier extraction	**46.20**	+63.1%
TQ outlier (baseline)	39.73	—
KIVI	470.50	—

*v2a variant: Hadamard without outlier handling hurts.

8.2 Fused Hadamard+Quantize Kernel

Steps 1–3 are fused into a single pass using AVX2 SIMD intrinsics (256-bit YMM registers, 4 f64 butterflies per cycle). The fused kernel avoids allocating the intermediate rotated tensor, achieving **1.6× speedup** over the separate-pass implementation at d=128 (benchmark: `benches/hadamard_simd.rs`).

8.3 Ablation: v1 vs v2 Components

Table 4 isolates the impact of each v2 change on Gemma 4 E2B 2-bit perplexity (R-α.1 protocol).

Key insight: Hadamard rotation alone (v2a) is worse than v1 because it spreads outliers without preserving them. The outlier extraction is the critical component—without it, the rotation amplifies quantization error on the high-kurtosis channels that transformers rely on for attention. The combination of Hadamard + outlier extraction (v2) achieves a 63% improvement over v1.

9 Cross-Architecture Perplexity Evaluation

We evaluate perplexity on WikiText-2 using the R-α.1 model surgery protocol (§3) across three models with fundamentally different KV-head architectures. Five methods are compared: FajarQuant v1, FajarQuant v2, KIVI, TurboQuant (with outlier handling), and the FP16 baseline.

Table 5: WikiText-2 perplexity (R-α.1 protocol, L=2048, n=30 chunks, 61,410 tokens). Lower is better. Bold = best quantized method per cell. Data source: `data/kv_cache/perplexity_v2_{gemma,mistral,qwen2}.json`.

Model	Bits	FP16	FQ v1	**FQ v2**	KIVI	TQ outlier
Gemma 4 E2B (MQA, 1 KV head)	2	28.13	125.19	46.20	470.50	**39.73**
	3	28.13	24.26	24.55	**21.90**	26.96
	4	28.13	26.75	26.98	35.17	27.33
Mistral 7B (GQA, 8 KV heads)	2	5.67	868.57	208.16	**23.96**	167.89
	3	5.67	19.44	9.52	**5.99**	9.48
	4	5.67	6.07	5.88	**5.73**	5.88
Qwen2-7B (GQA, 4 KV heads)	2	7.55	262.89	50.10	**46.70**	165.60
	3	7.55	8.93	8.21	**8.01**	8.26
	4	7.55	7.74	7.67	**7.62**	7.67

9.1 Architecture Dependence Is the Dominant Signal

The most striking finding is that **no single method wins everywhere**. KIVI wins 6 of 9 cells, but catastrophically fails on Gemma MQA at 2-bit (PPL 470, 16.7× FP16). This is because KIVI's per-channel symmetric quantization has no redundancy to exploit when there is only 1 KV head.

FajarQuant v2 is the **strongest non-KIVI method**, ranked second in 4 of 9 cells. On Qwen2-7B at 2-bit, FQ v2 achieves PPL 50.10 vs TQ outlier's 165.60 (**69.7% better**), approaching KIVI's 46.70. The "right" quantization method depends on the KV head count:

- **MQA (1 head):** Rotation-based methods essential. KIVI fails catastrophically.
- **Wide-GQA (≥8 heads):** KIVI's simplicity wins. Rotation adds overhead without benefit.
- **Narrow-GQA (4 heads):** In between. FQ v2 competitive with KIVI.

9.2 FQ v2 vs v1: The Upgrade Is Validated

FQ v2 improves over v1 in 7 of 9 cells, with massive gains at 2-bit: 63.1% on Gemma, 76.0% on Mistral, 80.9% on Qwen2. The calibrated PCA + outlier extraction redesign addresses the fundamental weakness of v1's naive per-tensor quantization.

10 Cross-Model Validation

To verify that FajarQuant's PCA rotation advantage is not specific to Gemma 4 E2B's MQA architecture, we extract real KV cache from two additional models with fundamentally different attention designs and re-run the 3-way comparison (Table 6).

Key finding: FajarQuant wins keys at all 3 bit widths on all 3 architectures, with single-digit-percent improvements that stay remarkably consistent across model families. The PCA-rotation advantage is *architecture-agnostic*: it generalizes from MQA (Gemma's 1 KV head shared across 8 attention heads) through GQA at both moderate (4:1, Mistral) and aggressive (7:1, Qwen2) sharing ratios. The smallest delta in the table (+3.6% for Mistral 2-bit vs TurboQuant at n=12,800) is well above any statistical-noise threshold and confirms the conclusion at scale.

10.1 When the PCA Value Gap Closes

A consistent pattern in the Gemma 4 E2B and Mistral 7B value-MSE results is that TurboQuant beats FajarQuant by 3–11% at every bit width on values, even though FajarQuant wins keys. This is the expected design tradeoff for any rotation method optimized on key distributions: PCA rotation aligns coordinates with the principal components of the *key* calibration buffer, which need not match the value distribution.

Qwen2-7B exhibits a regime change. On Qwen2-7B at 2-bit quantization, the FajarQuant-vs-TurboQuant value-MSE delta closes entirely to +0.3% (statistical tie at n=5,600), versus −2.8% on Gemma and −11.1% on Mistral at the same bit width. This is the first model in our evaluation where FajarQuant's value performance matches TurboQuant's at the most aggressive quantization level. The tie was observed at both n=560 (5-prompt dry run, +0.2%) and n=5,600 (50-prompt production, +0.3%), confirming it is not a small-sample artifact.

Table 6: Cross-architecture key MSE delta (% — positive = FajarQuant better, negative = baseline better). FajarQuant wins keys across all 3 architectures and all 3 bit widths (9 of 9 cells), confirming the PCA-rotation advantage is architecture-agnostic. Both Mistral 7B and Qwen2-7B data are from 50-prompt extractions: Mistral n=12,800 samples per cell, Qwen2-7B n=5,600. Gemma 4 E2B numbers are from Table 8 (n=1,500). The KV-head ratio column reports attention_heads : KV_heads, where 1:1 is full multi-head attention.

Model	Arch.	Layers	Heads (a:k)	σ_1/σ_2	Key MSE Δ vs TurboQuant		
					2-bit	3-bit	4-bit
Gemma 4 E2B	MQA	35	8:1	2.67	+**4.9**%	+**4.3**%	+**4.8**%
Mistral 7B	GQA	32	32:8	2.95	+**3.6**%	+**4.2**%	+**4.3**%
Qwen2-7B	GQA	28	28:4	**7.71**	+**4.7**%	+**5.4**%	+**5.4**%

The closure correlates with Qwen2-7B's exceptional key low-rank structure: its top-2 singular value ratio is $\sigma_1/\sigma_2 = 7.71$ in Table 6, versus 2.67 for Gemma and 2.95 for Mistral. We hypothesize that when keys are highly low-rank, the PCA rotation matrix that FajarQuant computes from the key calibration buffer also produces a useful rotation for the value path, because both share the same spatial directions in the residual stream of a transformer. On flatter-distribution models, the key-derived rotation is less informative for values and TurboQuant's data-agnostic random rotation wins.

This finding suggests two practical implications:

1. For models with strongly low-rank keys (high σ_1/σ_2), FajarQuant is competitive on *both* keys and 2-bit values—the value tradeoff documented elsewhere in this paper does not apply.
2. Future work could compute a separate value-distribution PCA rotation cache, decoupling the value-side rotation from the key calibration. Preliminary experiments are out of scope for this paper, but the Qwen2-7B result suggests the upper bound is at least "competitive with TurboQuant on values."

11 KV Cache Structure Analysis

We analyze the singular value distribution of real KV cache from Gemma 4 E2B (Table 7).

Real KV cache keys exhibit 2.45× stronger low-rank structure than synthetic random data. The structure varies significantly across layers: Layer 1 shows the strongest low-rank pattern ($\sigma_1/\sigma_2 = 4.04$), while Layer 9 shows the most uniform structure ($\sigma_1/\sigma_2 = 2.04$). This per-layer variation motivates adaptive per-head calibration.

Table 7: KV cache structure: real Gemma 4 vs synthetic

Metric	Real Gemma 4	Synthetic
Key σ_1/σ_2 ratio	2.73 ± 0.56	1.11 ± 0.07
Key top-3 energy	66.7%	99.7%
Value σ_1/σ_2 ratio	1.69 ± 0.31	—
Per-layer range	2.04–4.04	—
Samples	150	50

12 Ablation Study

We isolate the contribution of each innovation on real Gemma 4 E2B key data (150 samples, 10 prompts). Table 8 shows incremental gains as innovations are added.

Key findings: (1) PCA rotation provides 4–5% MSE improvement across all bit widths (4.9% at 2-bit, 4.3% at 3-bit, 4.8% at 4-bit), confirming that exploiting low-rank structure in KV cache is consistently beneficial. (2) Fused attention preserves identical MSE while reducing dequantization memory from $O(N \cdot d)$ to $O(2^b)$—a 524,288× reduction at N=16K. (3) Hierarchical allocation saves 48.7% of bit budget at N=10K by assigning 4 bits to the 256 most recent tokens and 1 bit to tokens beyond position 4096. The savings increase with context length (55.7% at N=16K).

Notably, "no rotation" (equivalent to KIVI per-channel) is 1.5% *worse* than random rotation at 2-bit, confirming that rotation-based approaches—whether random or adaptive—are fundamentally superior for KV cache quantization.

Table 8: Ablation: incremental innovation contributions (Gemma 4 E2B, N=10K)

Configuration	Key MSE (b=2)	MSE Δ vs TurboQuant	Dequant Memory	Bit Savings
No rotation (KIVI-like)	9.31×10^{-4}	+1.5%	$O(N \cdot d)$	0%
Random rotation (TurboQuant)	9.18×10^{-4}	—	$O(N \cdot d)$	0%
+ PCA rotation	$\mathbf{8.72 \times 10^{-4}}$	−4.9%	$O(N \cdot d)$	0%
+ Fused attention	$\mathbf{8.72 \times 10^{-4}}$	−4.9%	$O(2^b)$	0%
+ Hierarchical (Full FajarQuant)	$\mathbf{8.72 \times 10^{-4}}$	−4.9%	$O(2^b)$	48.7%

Table 9: End-to-end summary

Contribution	Metric	Result
Cross-arch analysis	Models tested	3 (MQA+GQA)
FQ v2 vs v1 (2-bit)	PPL improvement	63–81%
FQ v2 vs TQ (Qwen2)	PPL at 2-bit	**−69.7%**
Fused (N=16K)	Mem. saved	4 MB→64 B
Hierarchical	Bit savings	**48.7%**
Native impl	vs Python	**5.0**× faster
Safety	Overhead	Zero

Table 10: End-to-end memory (L=32, H=32, d=128)

N	FP16	Flat 3-bit	Hier.	Ratio
1K	512 MB	64 MB	35 MB	14.6×
4K	2 GB	256 MB	140 MB	14.6×
16K	8 GB	1 GB	547 MB	14.6×
64K	32 GB	4 GB	2.19 GB	14.6×

13 End-to-End Results

13.1 Memory Budget

14 Related Work

14.1 Per-Channel / Per-Token Methods

KIVI [Liu et al., 2024] observes that keys have large magnitude variation across channels and values across tokens, using per-channel key and per-token value quantization. Our cross-architecture evaluation reveals KIVI is the dominant method on wide-GQA models but fails on MQA (Table 5).

KVQuant [?] extends per-channel quantization with non-uniform datatypes and dense-and-sparse decomposition. We adopt their model-surgery evaluation protocol (R-α.1, §3).

SKVQ [?] combines per-channel sliding window with outlier handling. We cite their non-overlapping chunk evaluation protocol alongside KVQuant as our R-α.1 sources.

14.2 Rotation-Based Methods

TurboQuant [Zandieh et al., 2025] applies random orthogonal rotation followed by coordinate-wise Lloyd-Max quantization, achieving 2.7×-optimal MSE. FajarQuant v2 replaces the random rotation with calibrated PCA + outlier extraction.

SpinQuant [?] learns a rotation matrix end-to-end via Cayley optimization, targeting weight+activation quantization. FajarQuant uses calibration-derived PCA instead of learned rotation, avoiding the training cost.

QuIP# [Tseng et al., 2024] uses Hadamard incoherence processing for weight quantization with lattice codebooks. FajarQuant v2 borrows the Hadamard transform for outlier spreading but applies it to KV cache, not weights.

FlatQuant [?] flattens weight distributions via learnable affine transforms before quantization. The "flatten before quantize" principle is shared with FajarQuant's PCA rotation.

RotateKV [?] applies Hadamard rotation specifically to KV cache to reduce outlier magnitudes. FajarQuant v2 adds calibrated PCA on top of the Hadamard transform.

14.3 Adaptive Allocation Methods

KVTC [?] compresses KV cache via learned token-level compression. FajarQuant's hierarchical allocation achieves a similar goal (reducing budget for old tokens) without learned compression, using fixed bit schedules.

KVLinC [?] linearizes attention for efficient compressed KV cache. Complementary to quantization.

AsymKV [?] uses asymmetric quantization for keys vs values. Our results support this direction: the "right" quantization differs between keys and values (§10.1).

14.4 Weight Quantization (Complementary)

AQLM [Egiazarian et al., 2024]: Additive quantization for model weights. **FlexGen** [Sheng et al., 2023]: Memory offloading system. Both are complementary to KV cache quantization.

14.5 Novelty of This Work

No prior work provides a systematic cross-architecture perplexity comparison of KV cache quantization methods across MQA, wide-GQA, and narrow-GQA. Our finding that no single method dominates has not been documented previously. FajarQuant v2's combination of calibrated PCA, outlier extraction, fused Hadamard+quantize, and native language-level implementation is unique.

15 Performance Evaluation

The native Fajar Lang implementation (§7) runs **5.0× faster** than the Python+NumPy reference on an 8-token × 64-dim workload (28 ms vs 142 ms, release build, 10 iterations averaged). The fused Hadamard+quantize kernel (§5) contributes a 1.6× speedup over separate passes via AVX2 SIMD intrinsics.

Full per-layer latency profiling, inference throughput, and peak memory characterization across KIVI, TurboQuant, and FajarQuant are planned as future work. The benchmark methodology is locked in `bench/METHODOLOGY.md`: criterion with 100 samples, median ± 95% CI reporting, CPU governor pinned to `performance`, Turbo Boost disabled, and HT siblings offlined. The noise floor on our hardware (i9-14900HX, RTX 4090 Laptop) is 0.720 ns (`bench/results/noise_floor.json`).

16 FajarQuant v3.1: Adaptive Per-Head Selection

The cross-architecture analysis in §9 establishes that no single method wins everywhere. This motivates an **adaptive system** that selects the optimal quantization strategy *per head* based on each head's statistical profile.

16.1 Method

An initial prototype (v3.0) used MSE-based selection only, which led to catastrophic failures at 2-bit where MSE and PPL diverge (§16.2). FajarQuant v3.1 addresses this with a three-stage pipeline:

1. **Per-head profiling.** For each KV head, compute channel coefficient of variation (CV), kurtosis, and SVD ratio from a calibration buffer. These statistics determine whether the head's distribution favors per-channel quantization (KIVI-like) or rotation-based methods (PCA, Hadamard+outlier).
2. **Architecture-aware strategy selection.** An MSE-based threshold tuner assigns each head to one of five paths: A (KIVI-style per-channel), B (calibrated PCA rotation), C (Hadamard+outlier, i.e., TurboQuant-style), D (residual quantization), or E (asymmetric key/value bit allocation). In practice, the tuner routes predominantly to paths A–C; paths D and E are available but were not selected in our 3-model evaluation. For 2-bit, a PPL-guided fallback overrides the MSE-based assignment, because MSE and PPL diverge at extreme quantization (§16.2).
3. **Per-head PCA calibration.** For GQA models with multiple KV heads, PCA rotations are computed per-head rather than per-layer. This avoids cross-head contamination where heads with different distributions share a single rotation matrix.

16.2 PPL–MSE Divergence at 2-Bit

At 2-bit (4 quantization levels), rotation-based methods can achieve lower per-head MSE but higher end-to-end PPL, because systematic rounding errors compound through multi-layer attention. The MSE-based tuner assigns Path B (PCA) to heads where it reduces local error, but the accumulated global error increases perplexity. FajarQuant v3.0 suffered catastrophic failures from this effect: Gemma 2-bit PPL 171 and Mistral 2-bit PPL 79. V3.1's PPL-guided fallback tests the top-3 candidate strategies on a small PPL sample (5 chunks) and picks the lowest, eliminating all catastrophic failures.

16.3 Results

Table 11 shows v3.1 results across all 9 evaluation cells using the same R-α.1 protocol as §9.

16.4 Analysis

Score: 2 wins, 5 ties, 2 losses. V3.1 matches or beats the best fixed method in 7 of 9 cells. The

Table 11: FajarQuant v3.1 adaptive selection vs fixed methods (R-α.1 protocol, WikiText-2 test). Bold = best quantized method per cell. "v3.1 Strategy" shows the per-head path assignment. Data source: `data/kv_cache/perplexity_v3.1_*.json`.

Model	Bits	FP16	**FQ v3.1**	KIVI	TQ outlier	v3.1 Strategy
Gemma 4 E2B (MQA, 1 KV head)	2	28.11	39.91	480.66	**39.91**	PPL-guided → Path C
	3	28.11	**16.51**	21.69	26.40	17A + 13B
	4	28.11	28.13	35.11	**27.52**	17A + 13B
Mistral 7B (GQA, 8 KV heads)	2	5.67	**24.95**	**24.95**	163.96	PPL-guided → all-A
	3	5.67	6.32	**6.00**	9.44	278A + 234B
	4	5.67	**5.73**	**5.73**	5.88	all-A
Qwen2-7B (GQA, 4 KV heads)	2	7.69	**18.44**	28.53	75.15	PPL-guided
	3	7.69	8.15	**8.14**	8.38	222A + 2B
	4	7.69	**7.78**	**7.78**	7.82	all-A

2 losses are marginal: Gemma 4-bit (+2.2% vs TQ) and Mistral 3-bit (+5.3% vs KIVI).

Catastrophic failure elimination. V3.0 (MSE-only selection) produced 2 catastrophic cells: Gemma 2-bit PPL 171 and Mistral 2-bit PPL 79. V3.1's PPL-guided fallback reduces these to 39.91 and 24.95 respectively—matching the best fixed method in both cases.

Architecture-specific discoveries. The adaptive system finds two optima that no fixed method achieves:

- **MQA at 3-bit:** PCA rotation beats KIVI by 24% on Gemma (PPL 16.51 vs 21.69). The single KV head concentrates all information, making PCA's decorrelation highly effective.
- **GQA at 2-bit:** PPL-guided selection beats KIVI by 35% on Qwen2 (PPL 18.44 vs 28.53). The strategy the selector finds is neither pure KIVI nor pure TQ—it is a novel per-head mixture that exploits Qwen2's 4-head structure.

Per-head PCA: MSE gains ≠ PPL gains for GQA. On Mistral 3-bit, per-head PCA calibration (v3.1) routes 46% of heads to Path B, improving per-head MSE by 7.4% over per-layer PCA (v3.0). However, PPL only improves by 0.06 (6.38 → 6.32), still losing to KIVI (6.00). This confirms that for wide-GQA, the redundancy across 8 heads makes KIVI's simpler per-channel approach sufficient.

KIVI dominance at 3–4 bit confirmed. For all three architectures at 3-bit and 4-bit, the adaptive selector routes to KIVI-dominant strategies (mostly or entirely Path A). The system confirms KIVI's optimality rather than competing with it—a finding that strengthens the "no single method wins" thesis from a different angle.

Table 12: v3.0 → v3.1 ablation: PPL per cell. Bold = improved. † = catastrophic failure in v3.0 (PPL > 50).

Model	Bits	v3.0	**v3.1**	Fix applied
Gemma	2	171.19†	**39.91**	PPL-guided
Gemma	3	25.17	**16.51**	Fixed patching
Gemma	4	**27.31**	28.13	(regression)
Mistral	2	78.69†	**24.95**	PPL-guided
Mistral	3	6.38	**6.32**	Per-head PCA
Mistral	4	5.76	**5.73**	Per-head PCA
Qwen2	2	27.88	**18.44**	PPL-guided
Qwen2	3	**8.13**	8.15	(noise)
Qwen2	4	**7.77**	7.78	(noise)

16.5 Ablation: v3.0 vs v3.1

Table 12 isolates the impact of each v3.1 fix by comparing against v3.0 (MSE-only selection, per-layer PCA, no PPL-guided fallback).

Key takeaways: (1) PPL-guided 2-bit fallback is the most impactful fix, eliminating both catastrophic failures (−131 PPL on Gemma, −54 on Mistral) and discovering a novel optimum on Qwen2 (−9.4 PPL). (2) Per-head PCA calibration provides marginal but consistent improvements on Mistral GQA (−0.03 to −0.06 PPL). (3) Two cells show noise-level regressions (<0.02 PPL) that are within measurement variance. (4) Gemma 4-bit regresses by 0.82 PPL; this is the cost of using the same strategy (17A+13B) that was optimized for 3-bit—a future per-bit strategy could recover this.

17 Broader Impact

Aggressive KV cache quantization enables LLM deployment on resource-constrained devices (drones, medical instruments, automotive ECUs) that would otherwise require cloud connectivity. This has positive implications for latency-critical applications, data privacy (on-device inference avoids transmitting sensitive inputs), and accessibility (lower hardware cost broadens who can deploy LLMs).

However, quantization degrades model quality in ways that are input-dependent and difficult to predict. At 2-bit, perplexity increases 3–7× over FP16 even with adaptive selection (Table 11). For safety-critical applications, the gap between "works on WikiText-2" and "works on the deployment distribution" must be validated per deployment. We recommend that any production deployment of sub-4-bit KV cache quantization include distribution-specific perplexity monitoring and automatic fallback to higher precision when quality degrades beyond a user-defined threshold.

Our adaptive selector partially mitigates this risk by eliminating catastrophic failures (PPL $> 50\times$ FP16), but it does not guarantee bounded degradation on arbitrary inputs. The PPL-guided fallback (§16.2) is a step toward runtime safety but currently operates offline during calibration, not during inference.

18 Conclusion

We present the first systematic cross-architecture evaluation of KV cache quantization methods, testing across three fundamentally different attention designs (MQA, wide-GQA, narrow-GQA). Our primary finding is that **no single method dominates**: KIVI wins 6/9 cells in the fixed-method evaluation but catastrophically fails on MQA at 2-bit.

FajarQuant v3.1 addresses this with **adaptive per-head method selection**. By profiling each KV head and routing to the optimal quantization strategy, v3.1 matches or beats the best fixed method in 7/9 cells with zero catastrophic failures. The system discovers architecture-specific optima—PCA rotation for MQA at 3-bit (-24% vs KIVI) and PPL-guided selection for GQA at 2-bit (-35% vs KIVI)—while confirming KIVI's dominance at 3–4 bit GQA. The adaptive system's value lies not in universally beating any single method, but in *never catastrophically failing* and *automatically discovering* when alternatives outperform the default.

Combined with fused Hadamard+quantize (1.6× speedup), hierarchical bit allocation (48.7% savings), per-head PCA calibration, and a native Fajar Lang implementation (5× faster than Python), the system enables practical embedded deployment with compile-time safety guarantees.

Future Work

1. Separate value-distribution PCA rotation (Qwen2-7B results suggest this closes the value gap).
2. Online PCA update during autoregressive generation.
3. Learned rotation (SpinQuant-style) as a sixth path in the adaptive selector.
4. Extension from KV cache to model weight quantization.
5. Evaluation on larger models (Gemma 4 31B, Llama 3).

A Reproducibility

Listing 3: Reproducing all results

```
git clone https://github.com/fajarkraton/
    fajar-lang
cd fajar-lang && cargo build --release

# Rust tests (23 FajarQuant tests)
cargo test --lib -- runtime::ml::fajarquant

# KV cache extraction (requires GPU + PyTorch
    )
python scripts/extract_kv_cache.py \
  --model google/gemma-4-E2B --num-prompts 50

# 3-way MSE comparison
python scripts/run_comparison.py \
  --data data/kv_cache/ --num-prompts 10

# Perplexity evaluation
python scripts/eval_perplexity.py \
  --model google/gemma-4-E2B --bits 2,3,4

# Ablation study
python scripts/run_ablation.py \
  --data data/kv_cache/ --num-prompts 10
```

Environment: NVIDIA RTX 4090 Laptop GPU (16 GB), Intel i9-14900HX, 32 GB RAM, Ubuntu 24.04, Rust 1.87, PyTorch 2.11+cu130, Transformers 5.5.

Model: Google Gemma 4 E2B (5B parameters, 35 layers, 1 KV head, d=256, Apache 2.0 license).

Dataset: WikiText-2-raw-v1 [Merity et al., 2017] test split (1.29M characters).

B Proof of Theorem 3

Proof. The MSE of per-coordinate quantization after rotation R is $\mathrm{MSE}(R) = \sum_i D_i(\sigma_i^2, b)$ where σ_i^2 is the variance of coordinate i after rotation, and D_i is the per-coordinate distortion.

For PCA rotation R_{pca}, coordinates are principal components with ordered variances $\lambda_1 \geq \cdots \geq \lambda_d$. Per-coordinate quantization allocates independent scale/zero-point per dimension, adapting to each component's dynamic range.

For random rotation R_{rand}, coordinate variances are approximately equalized at $\mathrm{tr}(C)/d$.

With per-coordinate quantization, distortion $D_i \propto \mathrm{range}_i^2/4^b$. PCA minimizes total range by concentrating variance: low-variance components have small range and low distortion. The improvement is:

$$\frac{\mathrm{MSE}(R_{\mathrm{pca}})}{\mathrm{MSE}(R_{\mathrm{rand}})} = \frac{\sum_i \lambda_i^2}{\left(\sum_i \lambda_i\right)^2 / d} \leq 1 \tag{5}$$

with equality iff all eigenvalues are identical (no structure to exploit). □

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